

1. Solve following questions **using Stream API, you should write complete code context** rather than a single Stream API statement (please also write necessary POJO classes or helper code):
 1. Find top 3 longest strings that start with a vowel
 2. Return names of departments where average employee salary > 100,000
 3. Get a sorted list of all unique tags from a list of blog posts (Blog to Tags is 1-to-many relationship)
 4. Return top 5 words by frequency from a paragraph
 5. Group products by category and sort each group by price descending
2. Write code snippet to explain how **Optional** helps prevent null pointer exception, you may use blog-tags, and product-category POJOs to demo.
3. Explain why Java Stream API is required, how does it help on data processing?